CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 31

2 – BIT HALF ADDER USING NOR GATES

AIM: To design and implement a 2 – Bit Half Adder using NOR gates circuit using Logisim simulator.

ALGORITHM

1. Open the Logisim simulator.
2. Take four inputs: A1, A0, B1, B0.
3. Design a 1-bit Half Adder using NOR gates.
4. Use the same Half Adder circuit for both bit positions.
5. Connect A0 and B0 to the first Half Adder.
6. Connect A1 and B1 to the second Half Adder.
7. Obtain outputs S0, S1 and carries C0, C1.
8. Connect the outputs to LEDs/Output Pins.
9. Test the circuit with different input combinations.
10. Verify the outputs.

Truth Table:

A1	A0	B1	B0	C1	S1	S0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	0	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1

1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

Inputs: A1 A0, B1 B0
Outputs: C1 S1 S0

Here S = A + B and C1 is the final carry.

RESULT: